MILKY WAY AND OTHER GALAXIES

The Local Group

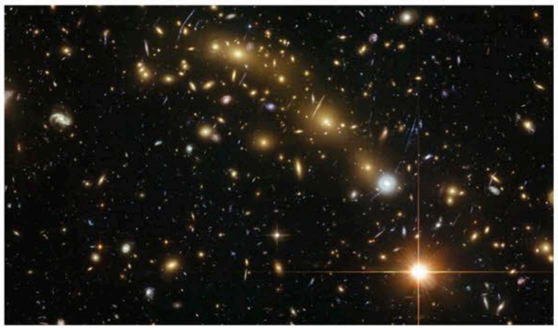
The Local Group of galaxies is a gravitationally bound cluster of over 54 galaxies, mainly dwarfs, and is about 10 million light-years across. It is dominated by three giant galaxies; the Milky Way, Andromeda, and Triangulum. Each of these has a swarm of orbiting smaller satellite galaxies. The Local Group was first recognized in 1936 by the American astronomer Edwin Hubble. The membership of some of the outliers (such as Antlia Dwarf, Sextans A, and NGC 3109) is debatable. Other, as yet undiscovered, members could be hidden behind the giant galaxies.

THE LOCAL GROUP OF GALAXIES

Name	Type	Distance from Solar System light-years	Diameter	Name	Type	Distance from Solar System light-years	Diameter
Milky Way	Barred spiral	0	100,000	IC 1613	Irregular	2,365,000	10,000
Sagittarius Dwarf	Dwarf elliptical	78,000	20,000	NGC 147	Dwarf elliptical	2,370,000	10,000
Ursa Major II	Dwarf elliptical	100,000	1,000	Andromeda III	Dwarf elliptical	2,450,000	3,000
Large Magellanic Cloud	Disrupted barred spiral	165,000	25,000	Cetus Dwarf	Dwarf elliptical	2,485,000	3,000
Small Magellanic Cloud	Irregular	195,000	15,000	Andromeda I	Dwarf elliptical	2,520,000	2,000
Boötes Dwarf	Dwarf elliptical	197,000	2,000	LGS 3	Irregular	2,520,000	2,000
Ursa Minor Dwarf	Dwarf elliptical	215,000	2,000	Andromeda Galaxy (M31)	Barred spiral	2,560,000	140,000
Sculptor Dwarf	Dwarf elliptical	258,000	3,000	M32	Dwarf elliptical	2,625,000	8,000
Draco Dwarf	Dwarf elliptical	267,000	2,000	M110	Dwarf elliptical	2,960,000	15,000
Sextans Dwarf	Dwarf elliptical	280,000	3,000	IC 10	Irregular	2,960,000	8,000
Ursa Major I	Dwarf elliptical	325,000	3,000	Triangulum Galaxy (M33)	Spiral	2,735,000	55,000
Carina Dwarf	Dwarf elliptical	329,000	2,000	Tucana Dwarf	Dwarf elliptical	2,870,000	2,000
Fornax Dwarf	Dwarf elliptical	450,000	5,000	Pegasus Dwarf	Irregular	3,000,000	6,000
Leo II	Dwarf elliptical	669,000	3,000	WLM	Irregular	3,020,000	10,000
Leo I	Dwarf elliptical	815,000	3,000	Aquarius Dwarf	Irregular	3,345,000	3,000
Phoenix Dwarf	Irregular	1,450,000	2,000	SAGDIG	Irregular	3,460,000	3,000
NGC 6822	Irregular	1,520,000	8,000	Antlia Dwarf	Dwarf elliptical	4,030,000	3,000
NGC 185	Dwarf elliptical	2,010,000	8,000	NCG 3109	Irregular	4,075,000	25,000
Andromeda II	Dwarf elliptical	2,165,000	3,000	Sextans A	Irregular	4,350,000	10,000
Leo A	Irregular	2,250,000	4,000	Sextans B	Irregular	4,385,000	8,000

Galaxy clusters and groups

Galaxies in the Universe are not distributed at random. They are in gravitationally bound groups containing tens to thousands of individuals. Dominating our region is The Great Attractor (the main component of which is the Norma Cluster). This is so massive it affects the normal expansion of the Universe discovered by Edwin Hubble. Clusters accumulate together to form superclusters. Cluster diameters are between 6 and 30 million light-years.



Galaxy cluster MACS J0416.1–2403 in Eridanus

GALAXY CLUSTERS AND GROUPS

Name	Distance millions of light-years	Recessional velocity miles per second (km per second)
Local Group	0	
M81 Group	11	207 (334)
Centaurus Group	12	186 (299)
Sculptor Group	12.7	181 (292)
Canes Venatici I Group	13	300 (483)
Canes Venatici II Group	26	387 (703)
M51 Group	31	345 (555)
Leo Triplet	35	386 (662)
Leo I Group	38	423 (680)
Draco Group	40	437 (704)
Ursa Major Group	55	631 (1,016)
Virgo Cluster	59	708 (1,139)